
Detection, Contextualization, and Explanation of Misinformation and Disinformation Using Cost-Effective Large Language Models

Ian Gotcher

Anuj Sakarda

Joshua Lu

Hai Yen Tran

1 Motivation

With so much happening in the world, information about what is happening is constantly flooding into the internet. With all this information, comes misinformation and disinformation, which is always being spread across social media, blogs, tabloids, and group chats. This misinformation can and has swayed opinions and led to societal harm, and the prospect of creating and spreading disinformation has opened the door for determined state adversaries to utilize large networks of bot accounts to manipulate the conversation.

The problem of how to address misinformation has been a topic of interest for academic research and a headache for large technology companies, politicians, and journalists. Various approaches have been tried, for example, fact-checking social media posts using reliable and well-known news agencies such as Reuters and The Associated Press was long a popular approach, but this approach has largely been phased out (many would say due to company takeovers or governmental pressure) in favor of community fact-checking. Both of these approaches have their limitations: news agency fact-checking can be expensive and limited to the most newsworthy information; while community fact-checking relies upon the work of volunteer contributors and the trustworthiness of the community members.

2 Related Work

Misinformation detection has been approached in several ways. The most traditional approach involves professional fact-checkers at organizations like PolitiFact, FactCheck.org, and Snopes, sometimes in partnership with news agencies like Reuters and the Associated Press. While these produce high-quality verdicts, they are expensive and can only cover a small fraction of the misinformation that circulates online.

Platforms have also experimented with crowdsourced fact-checking. Twitter’s Community Notes program allows volunteer contributors to append corrections to posts, using a bridging algorithm to surface notes that find consensus across users with differing viewpoints [1]. While crowdsourced corrections can be effective, studies have shown that the majority of proposed corrections on political content never reach the threshold needed to be publicly displayed [2].

More recently, researchers have explored using LLMs to automate the fact-checking process. These systems typically prompt a model to classify a claim as true, false, or somewhere in between, and often ask the model to explain its reasoning as well [3]. This is fast and flexible, but LLMs are still prone to hallucinate and confidently claim things that are wrong, which is obviously a very serious concern [4].

Our system builds on this line of work, using Gemini 3.1 Flash-Lite to classify and explain user-submitted claims through a simple GUI. We extend the typical verdict schema with "Needs Context" and "Unknowable" categories to better handle claims that are not straightforwardly true or false.

3 Proposed Approach and System Design

Large Language Models (LLMs) and the chatbots and APIs they power have become increasingly sophisticated, reliable, and able to work with live information within the past few years. Furthermore, the smallest, fastest, and least-expensive models have also experienced significant improvements on benchmarks. For example, Google's most recent smallest model Gemini 3.1 Flash-Lite scored an 86.9% on the GPQA Diamond benchmark, completely eclipsing the score of Google's earlier frontier models, with Gemini 1.0 Ultra scoring just 36%, and even Gemini 2.5 Pro (from just under a year ago) scoring 84%.

We leverage a fast and cost-effective LLM to create a system that can determine the veracity of a given statement within seconds while simultaneously generating an explanation of the context surrounding the statement, especially for more complex statements that are not black-and-white.

We designed and implemented a user-friendly GUI application that allows users to enter any statement that they want evaluated for truthfulness. Once the user enters their statement, their statement is sent to the Google Gemini 3.1 Flash-Lite API. The model has been specially instructed to evaluate the statement, explain its truthfulness, and, at the end of its response, return a verdict of True, False, Needs Context, or Unknown. The program processes this response using regular expressions to determine which verdict the model returned, and changes the truthfulness indicator to reflect the result of the evaluation.

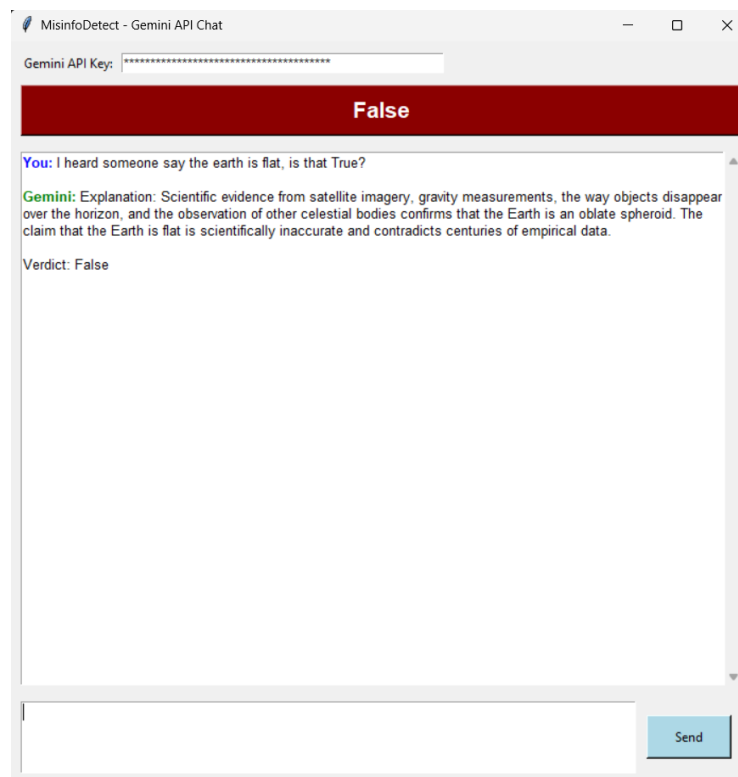


Figure 1: Screenshot of the Misinformation Detection Program in Use.

4 Results

We evaluated our system on a benchmark dataset of 32 statements covering all target verdict categories. In addition to label correctness, we also evaluated whether the model consistently followed the required output format and whether the GUI successfully extracted the returned verdict. This evaluation was intended as a prototype-level benchmark rather than a large-scale standardized test. As shown in Table 1, the system showed strong overall performance across the 32 benchmark statements, with no major failures, meaning it never reversed clearly true and false claims. It also achieved perfect format compliance and processing reliability, always returning a valid verdict that the GUI successfully extracted.

Metric	Value
Total benchmark statements	32
Completely correct	24
Minor failures	8
Major failures	0
Major failure rate	0%
Prompt obedience rate	100%
Processing success rate	100%

Table 1: Benchmark evaluation summary

The error pattern was highly structured. One common failure mode was that clearly false statements were labeled as "Needs Context" when they included a small amount of true background information or an unusual edge case. For example, the statement about artificial sweeteners causing cancer, the claim that wearing a seatbelt is more dangerous, and the statement suggesting that grapes should be replaced with chocolate because of sugar content. In these cases, the model’s explanations often acknowledged the broader evidence against the claim, but the final verdict remained overly cautious.

A second recurring failure mode was confusion between "Needs Context" and "Unknowable". This appeared most often for subjective, definition-dependent, or style-dependent claims, such as whether LeBron James is the “undisputable GOAT,” whether a hot dog is a sandwich, whether the Oxford comma is better, and whether “aluminum” is the correct pronunciation over “aluminium.” These examples suggest that the system sometimes treated context-dependent or convention-based claims as if they were fundamentally unknowable.

One especially informative case was the statement claiming that hominids first made fire at a precise historical time and date. The correct label was "Unknowable", but the system returned "False". However, the explanation itself correctly argued that such a precise claim is impossible to verify. This indicates that some failures may come less from misunderstanding the statement and more from instability in the final label selection step.

Overall, these results suggest that the system is a strong prototype for first-pass claim screening. Its most encouraging property is the absence of major true/false reversals, together with perfect format compliance and successful verdict extraction. At the same time, the benchmark also shows that the system is not yet well calibrated for nuanced boundary cases, especially those involving subjective judgments, competing conventions, or statements that mix false conclusions with partially true background details. Because the benchmark is small and manually curated, these results should be interpreted as evidence of feasibility rather than as a definitive measure of real-world misinformation detection performance.

5 Discussion

We successfully built a functional and cost-effective prototype for a misinformation detection GUI using Gemini 3.1 Flash-Lite. The system’s first pass achieved a 100% prompt obedience rate and processing success rate, and had zero major failures where clear truths or falsehoods were reversed. These findings suggest that the system is a promising scalable alternative, or complement, to traditional human-based fact-checking frameworks as outlined in Related Work.

We hypothesize that the model’s tendency to use undue caution in labeling false statements as "Needs Context" stems from its underlying Reinforcement Learning from Human Feedback (RLHF). This

training penalizes definitive claims, such as a "False" verdict, to minimize the risk of hallucinations; as a result, the model prioritizes safer, more neutral output. This failure mode could be mitigated by implementing a Retrieval-Augmented Generation (RAG) pipeline into the GUI through a Web Search API. This acts as an external anchor beyond the model's static training data, allowing the system to conduct real-time fact checking. The empirical evidence returned could overcome the existing RLHF-induced tendency towards undue caution, allowing the model to provide more conclusive verdicts. Furthermore, a system containing RAG could resolve temporal dependencies, accurately identifying claims that were accurate at specific points in time but have since become false.

The system's confusion between "Needs Context" and "Unknowable" statements are attributable to its failure to distinguish between lack of empirical evidence and subjective ambiguity when faced with a lack of clear consensus in training data. A similar conceptual confusion resulted in instability in the final labeling step. To address these concerns, future work could incorporate more explicit Chain of Thought (CoT) prompting during the explanation phase. This would require the model to evaluate the statement against specific criteria—such as availability of records about the claim, or linguistic markers of opinion (phrases like "the best")—before selecting a final output, improving its accuracy.

Finally, expanding the benchmark dataset with more edge cases and composite claims, such as primarily false statements attached to tangential or improbable truths, will further refine the model's ability to focus on the primary substance of a statement.

Once these architectural and prompt-level refinements are implemented, future work should compare these results to those of alternative LLMs, such as LLaMA and ChatGPT, to establish a more general performance baseline and validate our current system's design.

References

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